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/* CIS CIS 665: SQL - Quick Review */

-- 1) List building name, status, and quality level.

select buildingname, status, qualitylevel
from building

-- 2) List all information about employees.

select * from employee

-- 3) What is the weekly pay (40 hour work week) for each electrician?

select employeeename, payrate * 40 as [Weekly Pay]
from employee
where skill = 'Electrical'

-- 4a) What are the highest and lowest pay rates?

select max(payrate) as [Highest Pay Rate], min(payrate) as [Lowest Pay Rate]
from employee

-- 4b) What is the average number of days that employees are assigned to building 4?

select avg(numdays) as [Average Number of Days Assigned to Building 4]
from assignment
where buildingfk = 4

-- 4c) What is the average weekly pay for all employees?

select avg(payrate * 40) as [Average Weekly Pay]
from employee

-- 4d) How many buildings have quality level 3?

select count(*) as [# of Buildings with QL 3] from building where qualitylevel = 3

-- 4e) If each employee worked 50 hours a week, what would be the total weekly payroll amount?

select sum(payrate * 50) as [Weekly Payroll]
from employee

-- 5a) List all employees whose skill begins with "Elec".

select employeeename from employee where skill like 'Elec%'

-- 5) List all employees who have the letter "l" or "o" as the second character in their skill.

select employeeename, skill from employee where skill like '_[lo]%'

-- 6a) List employees whose pay rate is less than $40.

select employeeename, payrate from employee where payrate < 40

-- 6b) List buildings that do not have a quality level of 3.

select buildingname, qualitylevel from building where qualitylevel != 3

-- 7) List all electricians and framers who make more than $37.

select employeeename, skill, payrate from employee where (skill = 'Electrical' or skill = 'Framing') and
    payrate > 37

-- 8) List employees whose pay rate is between $35 and $40.

select employeeename, payrate
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from employee
where payrate between 35 and 40

-- 9) List unique building numbers from the assignment table

select distinct buildingfk from assignment

-- 10) List the plumbers, roofers, and framers

select employeeename, skill from employee
where skill in ('Plumbing', 'Roofing', 'Framing')

-- 11) List employees alphabetically by skill, and alphabetically by name within each skill

select skill, employeeename
from employee
order by 1,2

-- 12a) For each type of building, what is the average quality level?

select buildingtype, avg(cast(qualitylevel as float)) as [Average Quality Level]
from building
group by buildingtype

-- 12b) For each type of building, what is the average quality level for buildings of status 1? Consider ✓
        only those types of buildings having a maximum quality level no higher than 3.

select buildingtype, avg(cast(qualitylevel as float)) as [Average Quality Level]
from building
where status =1
group by buildingtype
having max(qualitylevel) <= 3

-- 13a) Employee 3 has been assigned to start work on building 2 on 2/25/2019. Add this data to the ✓
        Assignment table.

insert into assignment(employeefk, buildingfk, startdate)
values(3,2,'2/25/2019');

-- 13b) b. Create a new table called OfficeBuilding. Add all Office buildings to this table.

select * into officebuilding
from building
where buildingtype = 'office';

-- 14) Delete all Office buildings from Building.

delete
from building
where buildingtype = 'office';

-- 15) Give painters a 5 percent pay rate increase.

update employee
set payrate = payrate * 1.05
where skill = 'Painting';

-- 16a) List employee names and their assignment information (BuildingFK, StarDate and NumDays).

select employeeename, buildingfk, startdate, numdays
from employee inner join assignment on employeeepk = employeeefk

-- 16a) Alternate

select employeeename, buildingfk, startdate, numdays
from employee, assignment
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where employeeek = employeeefk
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-- 16b) List the names of employees assigned to office buildings.
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select distinct employeeename as [Assigned to Office Buildings]
from employee inner join assignment on employeeek = employeeefk
inner join building on buildingfk = buildingpk
where buildingtype = 'Office'
```

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-- 16b) Alternate
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```
select distinct employeeename as [Assigned to Office Buildings]
from employee inner join assignment inner join building
on buildingfk = buildingpk
on employeeek = employeeefk
where buildingtype = 'Office'
```

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-- 17) List the names of employees assigned to office buildings.
```

```
select employeeename as [Assigned to Office Buildings]
from employee where employeeek in
(select distinct employeeefk
from assignment
where buildingfk in
(select buildingpk
from building
where buildingtype = 'office'));
```

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-- 18) List the names of employees and the names of their supervisors.
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select e.employeeename as [Employee], s.employeeename as [Supervisor]
from employee e inner join employee s
on e.supervisorfk = s.employeeek
```

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-- 18) Alternate (Outer Join)
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select e.employeeename as [Employee], isnull(s.employeeename,'Boss') as [Supervisor]
from employee e left outer join employee s
on e.supervisorfk = s.employeeek
```

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-- 19) List the names and pay rates of the employees who have the highest or lowest pay rates.
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select employeeename, payrate, 'Highest Pay' as [Pay Status]
from employee
where payrate =
(select max(payrate)
from employee)
union
select employeeename, payrate, 'Lowest Pay'
from employee
where payrate =
(select min(payrate)
from employee)
```

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-- 20) Which buildings are office buildings and have employee 3 assigned to them?
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```
select buildingname
from building
where buildingtype='Office'
intersect
select buildingname
from building
where buildingpk in
(select buildingfk
from assignment
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where employeefk =3)
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-- 21)List those employees who are assigned to more than 50 days of work.
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```
select employeename
from employee
where 50 <
(select sum(numdays)
from assignment
where employeepk = employeefk)
```

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-- 21) Alternate (uncorrelated)
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```
select employeename
from employee
where employeepk in
(select employeefk
from assignment
group by employeefk
having sum(numdays) > 50)
```

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-- 22) Which office buildings do not have employee 1 assigned to them?
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```
select buildingname
from building
where buildingtype = 'Office'
and not exists
(select *
from assignment
where buildingpk = buildingfk
and employeefk = 1)
```

```
--22) Alternate (uncorrelated)
```

```
select buildingname
from building
where buildingtype = 'Office'
and buildingpk not in
(select buildingfk
from assignment
where employeefk = 1)
```